9.6 Alternating Series and Conditional Convergence

A geometric series with a negative common ratio, the alternating harmonic series $\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n}$ and the alternating p-series $\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n^p}$ are examples of series with some negative terms.

To determine an infinite series with some negative terms is convergent or divergence, we may use the n-term test for divergence, the alternating series test or the test for absolute convergence.

Definition 1 (Alternating Series): A series whose terms are alternately positive and negative.

Example 1:

(a) $1-\frac{1}{2}+\frac{1}{3}-\frac{1}{4}+\frac{1}{5}-\cdots+\frac{(-1)^{n+1}}{n}+\cdots=\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n}$. **Alternating Harmonic Series** (Converges)

(b) $-2+1-\frac{1}{2}+\frac{1}{4}-\frac{1}{8}+\cdots+\frac{(-1)^{n}}{2^{n-2}}+\cdots\sum_{n=1}^{\infty}\frac{(-1)^{n}}{2^{n-2}}$. Geometric series with $|r|=\frac{1}{2}$ (Converges)

(c) $1-\frac{1}{\sqrt{2}}+\frac{1}{\sqrt{3}}-\frac{1}{\sqrt{4}}+\cdots+\frac{(-1)^{n-1}}{\sqrt{n}}+\cdots=\sum_{n=1}^{\infty}\frac{(-1)^{n-1}}{\sqrt{n}}$. **Alternating *p*-series** with $p=\frac{1}{2}$ (Converges)

(d) $1-2+3-4+5-6+\cdots+(-1)^{n+1}n+\cdots=\sum_{n=1}^{\infty}(-1)^{n+1}n$. Diverges by the nth-term test.

Theorem 1 (The Alternating Series Test):

The alternating series $\sum_{n=1}^{\infty}(-1)^{n+1}u_n$ converges if **all three** of the following conditions are satisfied:

(1) $u_n > 0 \quad \forall n$,

(2) $u_n \geq u_{n+1} \quad \forall n \geq N$, for some integer N, **and**

(3) $\lim_{n\to\infty} u_n = 0$.

Proof:

Assume $N = 1$, where we have nonincreasing terms $u_1 \geq u_2 \geq u_3 \geq \cdots$.

If n is an even integer, say $n = 2m$, then the sum of the first n terms is

$$s_{2m} = (u_1 - u_2) + (u_3 - u_4) + \cdots + (u_{2m-1} - u_{2m})$$

Written this way, we see that s_{2m} is the sum of n nonnegative terms, since each term in parentheses is positive or zero. Hence, $s_{2m+2} \geq s_{2m}$, and the sequence $\{s_{2m}\}$ is nondecreasing.

On the other hand, we can write

$$s_{2m} = u_1 - (u_2 - u_3) - (u_4 - u_5) - \cdots - (u_{2m-2} - u_{2m-1}) - u_{2m}$$

This shows that $s_{2m} \le u_1$ for every m. Hence the sequence $\{s_{2m}\}$ is both nondecreasing and bounded from above, so it has a limit, say $\lim_{m\to\infty} s_{2m} = L.$ ---------- (1)

If n is an odd integer, say $n = 2m+1$, then the sum of the first n terms is

$$s_{2m+1} = s_{2m} + u_{2m+1}$$

Since $\lim_{n\to\infty} u_n = 0,$ then $\lim_{m\to\infty} u_{2m+1} = 0,$ and

$$\lim_{m\to\infty} s_{2m+1} = \lim_{m\to\infty} s_{2m} - \lim_{m\to\infty} u_{2m+1} = L + 0 = L. \text{ ---------- (2)}$$

Combining the results of (1) and (2) gives $\lim_{n\to\infty} s_n = L.$ Hence, the series converges to L.

Example 2 (The Alternating Harmonic series):
Show that the alternating harmonic series $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$ converges.

Solution:
Let $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} = \sum_{n=1}^{\infty} (-1)^{n+1} u_n$ where $u_n = \frac{1}{n}$. Then,

(1) $u_n > 0 \ \ \forall n \ge 1$, (2) $\forall n \ge 1$, $\frac{1}{n} \ge \frac{1}{n+1} \Rightarrow u_n \ge u_{n+1},$ and (3) $\lim_{n\to\infty} u_n = \lim_{n\to\infty} \frac{1}{n} = 0$.

By the alternating series test, (1), (2) and (3), the alternating harmonic series converges.

Example 3: Does the series $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{\sqrt{n}+1}{n+1}$ converge?

Solution:

Let $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{\sqrt{n}+1}{n+1} = \sum_{n=1}^{\infty} (-1)^{n+1} u_n$ where $u_n = \frac{\sqrt{n}+1}{n+1}$. Then,

(1) $u_n > 0 \ \ \forall n \ge 1,$

(2) Let $f(x) = \frac{\sqrt{x}+1}{x+1}, \quad x \ge 1.$ Then $f'(x) = \frac{-(x + 2\sqrt{x} - 1)}{2\sqrt{x}(x+1)^2}$.

$$x + 2\sqrt{x} > 1 \quad \forall x \ge 1 \quad \Rightarrow \quad x + 2\sqrt{x} - 1 > 0$$

$$\Rightarrow f'(x) < 0 \quad (\because \ 2\sqrt{x}(x+1)^2 > 0 \quad \forall x \ge 1.) \ \Rightarrow \ f \text{ is decreasing}$$

$$\therefore \ u_n \ge u_{n+1} \quad \forall n \ge 1, \text{ and}$$

(3) $\lim_{n\to\infty} u_n = \lim_{n\to\infty} \frac{\sqrt{n}+1}{n+1} = \lim_{n\to\infty} \frac{\frac{1}{\sqrt{n}} + \frac{1}{n}}{1 + \frac{1}{n}} = 0.$

By the alternating series test, (1), (2) and (3) $\Rightarrow$ the series converges.

Theorem 2: (The Alternating Series Estimation Theorem)

If the alternating series $\sum_{n=1}^{\infty}(-1)^{n+1}u_n$ satisfies all the three conditions of Theorem 1,

then for $n \geq N$, $s_n = u_1 - u_2 + u_3 - u_4 + \cdots + (-1)^{n+1}u_n$

approximates the sum L of the series with an error whose absolute value is less than u_{n+1}, the absolute value of the first unused term.

Furthermore, the sum L lies between any two successive partial sums s_n and s_{n+1}, and the remainder, $L - s_n$, has the same sign as the first unused term.

Example 4:

Consider the geometric series $\sum_{n=0}^{\infty}\frac{(-1)^n}{2^n} = 1 - \frac{1}{2} + \frac{1}{4} - \frac{1}{8} - \frac{1}{16} - \frac{1}{32} - \frac{1}{64} - \frac{1}{128} + \frac{1}{256} - \cdots$

The sum of the first eight terms is $s_8 = 0.6640625$ and

the sum of the first nine terms is $s_9 = 0.66796875$.

The sum of the geometric series is $\frac{1}{1-(-1/2)} = \frac{2}{3} = S$ and

we note that $0.6640625 < 2/3 < 0.66796875$.

The difference, $S - s_8 = 2/3 - 0.6640625 = 0.0026041666$ is positive and is less than $|u_9| = 1/256 = 0.00390625$.

Example 5:

Determine how many terms should be used to estimate the sum of the series $\sum_{n=1}^{\infty}\frac{(-1)^n}{n^2+3}$ with an error of less than 0.001.

Solution:

$$\begin{aligned} |\text{Error}| < 0.001 &\Rightarrow |u_{n+1}| < 0.001 \\ &\Rightarrow \frac{1}{(n+1)^2+3} < 0.001 \\ &\Rightarrow (n+1)^2 + 3 > 0.001 \\ &\Rightarrow n > -1 + \sqrt{997} \approx 30.5753 \\ &\Rightarrow n \geq 31. \end{aligned}$$

Definition 2:

A series that is convergent but not absolutely convergent is called **conditionally convergent**.

When $\sum a_n$ converges but $\sum |a_n|$ diverges., we say $\sum a_n$ ***converges conditionally.***

Example 6 (The Alternating *p*-series):

The alternating *p*-series $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^p}$ converges absolutely for $p > 1$ and conditionally for $0 < p \le 1$.

Let $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^p} = \sum_{n=1}^{\infty} (-1)^{n-1} u_n$ where $u_n = \frac{1}{n^p}$ and $p > 0$. Then,

(1) $u_n > 0 \quad \forall n \ge 1$,

(2) When $p > 0$ and $\forall n \ge 1$, $\frac{1}{n^p} \ge \frac{1}{(n+1)^p} \Rightarrow u_n \ge u_{n+1}$, and

(3) When $p > 0$, $\lim_{n\to\infty} u_n = \lim_{n\to\infty} \frac{1}{n^p} = 0$.

By the alternating series test, (1), (2) and (3) , the alternating *p*-series $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^p}$ converges for $p > 0$.

$\sum_{n=1}^{\infty} \left| \frac{(-1)^{n-1}}{n^p} \right| = \sum_{n=1}^{\infty} \frac{1}{n^p}$ converges when $p > 1$ and diverges when $p \le 1$.

$\therefore$ The alternating *p*-series $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^p}$ converges absolutely when $p > 1$ and conditionally when $0 < p \le 1$.

<u>Strategy for Testing Series</u>

- No fast rules about which test to apply to a given series, but the following advice may help.
- It is not wise to apply a list of the tests in a specific order until one finally works. Instead, the main strategy is to classify the series according to its ***form***.

1. Identify the series when it is **geometric** or ***p*- series** (including the harmonic series).
2. If it is easily to show that $\lim_{n\to\infty} a_n \ne 0$ or limit does not exist, apply **the *n*th-term test** for divergence.
3. If $\int f(x)dx$ is easily evaluated, **the integral test.** (f is positive, continuous and decreasing $\forall x \ge N$.)
4. Series has a form ***similar*** to geometric or *p*- series, may use **the comparison test (direct/ limit)**.
   * a_n is a rational/ algebraic (involving roots of polynomial) function, should be compared with *p*- series.

5. Series that involve factorials or other product (including a constant raised to the *n* powers), may use **the ratio test**. Not suitable for rational/ algebraic because $\lim_{n\to\infty}\frac{a_{n+1}}{a_n}=1$.

6. a_n is of the form $(b_n)^n$, may use **the root test**.

7. a_n is of the form $(-1)^n b_n$ (an alternating series), series is non-increasing and $\lim_{n\to\infty} b_n = 0$, may use **the alternating test**.

8. $\sum a_n$ has some negative terms, test $\sum |a_n|$ for absolute convergence.

Example 7: Choose a suitable test for each of the following series:

(1) $\sum_{n=1}^{\infty}\frac{n+2}{2n-1}$ (2) $\sum_{n=1}^{\infty} ne^{-n^2}$ (3) $\sum_{n=1}^{\infty}\frac{(-1)^n}{n^3+2}$

(4) $\sum_{n=1}^{\infty}\frac{(-1)^n\sqrt{n^3+1}}{n^3+2}$ (5) $\sum_{n=1}^{\infty}\frac{5^n n!n!}{(5n)!}$ (6) $\sum_{n=1}^{\infty}\frac{n}{2^{(n^2)}}$

<u>Rearranging series</u>

Theorem 3: The Rearrangement Theorem for Absolutely Convergent Series
Suppose $\{b_n\}$ is an arrangement of $\{a_n\}$. If $\sum_{n=1}^{\infty} a_n$ **<u>converges absolutely</u>**, then $\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} a_n$.

Example 8: The series $1-\frac{1}{4}-\frac{1}{16}+\frac{1}{9}+\frac{1}{25}+\frac{1}{49}-\frac{1}{36}-\frac{1}{64}-\frac{1}{100}-\frac{1}{144}+\cdots$, is rearrangement of the terms of the *p*-series $1-\frac{1}{4}+\frac{1}{9}-\frac{1}{16}+\cdots+\frac{(-1)^{n+1}}{n^2}+\cdots$. Both series converge to the same sum since the *p*-series converges absolutely.

Remark: If $\sum_{n=1}^{\infty} a_n$ **<u>converges conditionally</u>**, $\sum_{n=1}^{\infty} b_n$ may converge to a **different sum** or may diverge.

Example 9: The alternating harmonic series $1-\frac{1}{2}+\frac{1}{3}-\frac{1}{4}+\cdots+\frac{(-1)^{n+1}}{n}+\cdots$ converges conditionally and it has sum ln 2. The series $1-\frac{1}{2}+\frac{1}{3}+\frac{1}{5}-\frac{1}{4}+\frac{1}{7}+\frac{1}{9}-\frac{1}{6}+\cdots$ is obtained from the alternating harmonic series by rearranged the terms but it has sum $\frac{3}{2}\ln 2$.